

IT 308 – ONLINE PERFORMANCE TASK

Title: “Innovating the Future: Emerging Technology Application Design”

Course Requirement Submission | Individual Output

- **Chosen Technology:** Artificial Intelligence (AI)
- **Supporting Technology:** Internet of Things (IoT)
- **Proposed App Title:** AgriSmart AI: An Artificial Intelligence-Based Smart Farming Assistant for Southern Leyte

I. Introduction

Agriculture remains one of the primary sources of income and livelihood in Southern Leyte and across the rural landscape of the Philippines. Despite its foundational importance, local farmers constantly face severe operational bottlenecks, including unpredictable weather cycles, sudden plant diseases, destructive pest infestations, and extremely limited access to on-site agricultural experts. In recent years, the escalating impacts of global warming and climate change have worsened these vulnerabilities. Rising global temperatures, erratic rainfall distributions, more frequent and intense super-typhoons, unexpected flooding, and prolonged dry spells directly threaten regional crop production.

These climate-driven anomalies completely disrupt traditional farming methods, making it nearly impossible for small-scale farmers to accurately plan planting calendars or manage irrigation cycles effectively. The consequence is a painful cycle

of massive crop loss, food insecurity, and deep financial instability for farming households. To directly mitigate these vulnerabilities, this concept paper proposes **AgriSmart AI**, a mobile-based smart farming assistant powered by Artificial Intelligence. This application is designed to deliver localized, data-driven insights, automated disease diagnostics, and predictive recommendations, empowering Southern Leyte's agricultural community to adapt to climate change, secure their harvests, and stabilize their economic yields.

II. Emerging Technology: Artificial Intelligence

Artificial Intelligence (AI) refers to a branch of computer science dedicated to building software systems capable of performing tasks that traditionally require human intelligence—such as learning, visual perception, reasoning, and automated decision-making. Instead of relying solely on static, pre-programmed code, AI systems process massive datasets, recognize complex underlying patterns, and autonomously adapt their outputs over time.

How it works in AgriSmart AI:

- **Data Collection:** The system aggregates localized telemetry data directly from the field. This includes user-uploaded crop images, real-time regional weather feeds, and granular environmental metrics like soil moisture and ambient temperatures.
- **Data Analysis:** Utilizing Computer Vision and Machine Learning models, the AI engine processes these inputs to instantly identify anomalies, such as early-

stage foliar diseases, nutrient deficiencies, or critical soil dehydration.

- **Pattern Recognition:** The AI continually correlates current farm data against extensive historical agronomic databases and regional climate trends. This allows it to identify hidden warning signs, such as predicting when specific weather combinations are highly likely to trigger pest outbreaks.
- **Actionable Recommendations:** Once the processing cycle is complete, the AI translates complex environmental metrics into direct, localized instructions. It generates automated alerts advising farmers on exact pest treatments, dynamic irrigation schedules, and optimized harvest windows.

III. Proposed Application: AgriSmart AI

AgriSmart AI is a highly accessible, specialized mobile application tailored explicitly to support the unique agricultural conditions of Southern Leyte. It transforms complex environmental variables into easy-to-read, actionable advice to optimize small-scale farming operations.

Key Features

- **AI Crop Disease Detection:** Farmers can capture and upload smartphone photos of infected or damaged crops. The embedded computer vision model scans the images, diagnoses the probable plant disease or pest type within seconds, and outputs an immediate

treatment plan using locally available, environmentally safe countermeasures.

- **Weather-Driven Farming Advice:** By integrating live climate forecasts and localized meteorological APIs, the app alerts farmers of incoming hazards (e.g., heavy rain, typhoons, or droughts) and modifies planting schedules to minimize risk.
- **Smart Irrigation Alerts:** The application processes real-time soil hydrology numbers to calculate precise crop water requirements, sending proactive push notifications to farmers to irrigate only when necessary, preventing both root rot from overwatering and crop stress from underwatering.
- **Predictive Crop Yield Forecasting:** Using a blend of historical farm data and seasonal weather tracking, the AI estimates the farmer's expected harvest volumes weeks in advance, allowing them to negotiate fair prices and coordinate logistics prior to cutting crops.
- **Hyper-Local Market Price Monitoring:** The system maintains a live database of current crop prices across local trading centers and public markets within Southern Leyte, protecting farmers from predatory middlemen and enabling data-driven selling strategies.

IV. Supporting Technology: Internet of Things (IoT)

To maximize the diagnostic accuracy of the core AI models, AgriSmart AI integrates the **Internet of Things (IoT)** as its primary supporting framework. IoT represents a system of physical objects embedded with sensors, processing power, and communication technologies that connect and exchange data with other systems over wireless networks.

In this architecture, low-cost hardware sensor nodes (measuring soil moisture, ambient temperature, and humidity) are deployed directly into the soil plots. These nodes transmit field data continuously to the central cloud platform. The integration of AI and IoT forms a highly effective symbiotic ecosystem: **IoT acts as the nervous system** that captures real-time, objective field ground truths, while **AI acts as the brain** that interprets that raw data into meaningful operational decisions. This combination eliminates human guesswork, prevents delayed responses to environmental threats, and establishes a highly responsive, climate-resilient farming ecosystem.

V. Target Users

AgriSmart AI is intentionally architected to support a diverse ecosystem of regional agricultural stakeholders:

- **Smallholder Farmers in Southern Leyte:** The primary end-users who rely on accessible, real-time field data and simplified visual alerts to protect daily crop yields.
- **Agricultural Cooperatives:** Group managers who need macro-level yield predictions and market trends

to effectively coordinate bulk sales, shared logistics, and storage requirements for their members.

- **Local Government Units (LGUs) & DA Offices:** Municipal agricultural officers who can utilize aggregated, anonymized regional farm data to deploy targeted government subsidies, disaster relief, and specialized farming programs.
- **Agricultural Students and Researchers:** Local state university researchers who can leverage the anonymized datasets to study changing climate behaviors and local crop resilience trends.

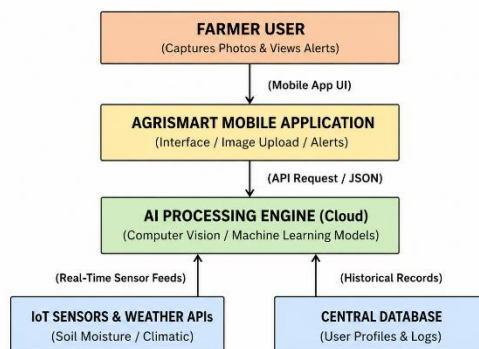
VI. Community Impact

The deployment of AgriSmart AI within Southern Leyte directly drives meaningful socio-economic advancements:

- **Mitigating Disaster-Driven Crop Loss:** Equips vulnerable rural communities with localized, early-warning alerts to actively protect investments before typhoons or pest outbreaks strike.
- **Fostering Climate Change Adaptation:** Transitions traditional farming from historical guesswork into an agile, data-driven methodology capable of navigating unpredictable global warming patterns.
- **Elevating Family Incomes:** Boosts the net profit of rural households by minimizing input waste (water, fertilizers, pesticides) and maximizing premium harvest yields.

- **Securing Local Food Supplies:** Stabilizes localized crop production cycles, reducing reliance on expensive food imports and safeguarding food security within the province.
- **Bridging the Rural Digital Divide:** Accelerates tech literacy among generational farmers by introducing approachable, highly beneficial digital utilities to their daily routines.

VII. System Architectural Design



VIII. Reflection

1. Why did you choose this technology?

I chose Artificial Intelligence because it possesses the unique capability to solve complex, non-linear real-world problems by analyzing vast data arrays in a manner that mimics human

expertise. For instance, a smallholder farmer struggling with a sudden, mysterious pest outbreak can utilize AI computer vision to instantly identify the threat and receive an accurate mitigation strategy without waiting for an on-site expert. AI excels at analyzing patterns within historical crop databases, making it an incredibly powerful tool for predicting vulnerabilities before they manifest as severe financial losses. I specifically focused on the agricultural sector because farming serves as the primary economic backbone for the majority of rural families in Southern Leyte. As climate change continues to trigger erratic rainfall patterns and unpredictable typhoons, traditional farming methods are becoming increasingly unreliable. AI can bridge this gap by interpreting live data to advise farmers on optimal planting windows, automated irrigation needs, and immediate disease diagnoses. Ultimately, I selected AI because it has the power to transform deep-seated agricultural challenges into viable opportunities for community development, financial security, and technological growth.

2. What challenges might occur in developing this?

One of the primary development challenges is acquiring a robust, highly accurate dataset of localized crop photos and soil profiles necessary to properly train the underlying machine learning models. If a farmer uploads a blurry, poorly lit smartphone photo, an unrefined AI engine could easily misidentify the plant disease, leading to incorrect treatment recommendations that might worsen crop damage. Furthermore, rural barangays across Southern Leyte often suffer from unstable cellular data connectivity, which poses a significant hurdle for real-time cloud computing and instant

alert dispatches. The financial cost associated with manufacturing and deploying physical, waterproof IoT sensor nodes across vast agricultural fields also presents a clear barrier to widespread adoption among low-income farmers. On a human level, older generational farmers who are accustomed to traditional practices may feel intimidated by a digital application or struggle to interpret automated alerts correctly. Overcoming these multi-faceted challenges will require rigorous physical prototyping, lightweight code optimization for low-bandwidth connectivity, and deep collaborative field training alongside local agricultural extension offices.

3. Is it realistic? Why or why not?

I believe that the AgriSmart AI framework is highly realistic because the underlying hardware and software technologies are already mature, accessible, and proven globally. Precision agriculture systems utilizing computer vision and automated irrigation have already demonstrated immense success in optimizing crop outputs in various developing nations. Rather than attempting an immediate, expensive province-wide rollout, the development can realistically begin as a localized pilot project targeting a single agricultural cooperative within a chosen barangay. By securing institutional backing from Local Government Units (LGUs) or academic research grants, the funding required for entry-level microcontrollers and sensor hardware can be effectively managed. Providing interactive, hands-on community workshops can smoothly guide farmers through the app's user interface, demystifying the technology. Implementing the system through a phase-by-phase rollout allows developers to monitor real-world

accuracy, gather user feedback, patch software bugs, and demonstrate clear financial benefits before scaling the infrastructure up to broader markets.

4. What skills must you improve to build this in the future?

To successfully transform this conceptual design into a fully functional, production-ready smart farming solution, I must aggressively improve my specialized technical competencies. First and foremost, I need to deepen my knowledge of Machine Learning frameworks, specifically learning how to train, optimize, and deploy convolutional neural networks (CNNs) for image classification and pest recognition. I also need to refine my mobile application development skills to ensure I can build an incredibly clean, responsive, and intuitive user interface that accommodates users with limited technical literacy. Furthermore, acquiring cloud data engineering skills will be vital to managing simultaneous, high-volume data streams coming from multiple IoT field devices. I must expand my hardware engineering foundations to properly configure, calibrate, and secure Arduino or Raspberry Pi microcontrollers against harsh environmental elements. Finally, building skills in data analysis, UX design, and cloud security will allow me to properly interpret system outputs, protect sensitive farmer demographics, and maintain system reliability under real, unpredictable field conditions.